TWO PROOFS OF THE CASSELS–SWINNERTON-DYER CONJECTURE FOR CUBIC SURFACES

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Abstract. The Cassels–Swinnerton-Dyer conjecture asserts that a cubic hypersurface contains a rational point if and only if it contains a point of degree coprime to 3, or, equivalently, a zero-cycle of degree 1. The case of smooth cubic surfaces in characteristic zero has been reduced by Coray and Voisin to the case of points of degree 4. We give two independent proofs of this missing case and use a lifting argument of Ma to extend the result to smooth cubic surfaces over arbitrary fields. We further give a separate argument for the case of singular cubic surfaces, extending previous work of Coray over perfect fields. Altogether, this proves the Cassels–Swinnerton-Dyer conjecture for cubic surfaces.

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1. Introduction

Cassels and Swinnerton-Dyer conjectured that a cubic surface with a point of degree coprime to 3, or, equivalently, a zero-cycle of degree one, has a rational point; see also the related problems of Segre [Seg51, p. 2]. The analogous criteria for quadrics and plane cubic curves are due to Springer [Spr52] and Poincaré [Poi01]; see [Cor76a, Proposition 2.3] for a modern treatment of the latter.

Coray formulated the Cassels–Swinnerton-Dyer conjecture for cubics of arbitrary dimension [Cor76a, Conjecture (CS)]. Over perfect fields, he proved the singular surface case [Cor76a, Proposition 8.2] and reduced the case of smooth cubic surfaces to points of degrees 4 and 10 [Cor76a, Theorem 7.1]. Further developments and related results appeared in [CT79, CTSD12, CT20, Ma21, RV22, Bal24, Por25]. Recently, Voisin [Voi26] showed that, in characteristic zero, a smooth cubic surface with a point of degree 10 has a rational point or a point of degree 4. This reduces the conjecture for smooth cubic surfaces in characteristic zero to the following statement, which we prove in this paper.

Theorem 1.1. *Let S be a smooth cubic surface over a field k of characteristic zero. If S has a closed point of degree 4, then it has a rational point.*

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In Section 5, we combine this result and the aforementioned work of Coray and Voisin with a lifting argument of Ma [Ma21] to deal with smooth cubic surfaces over arbitrary fields. Finally, in Section 6, we handle singular cubic surfaces over arbitrary fields by reducing to smooth cubic surfaces over $k((t))$, extending Coray's result over perfect fields. Together, these arguments give the following solution of the Cassels–Swinnerton-Dyer conjecture for surfaces; see Theorem 6.1 below for a geometric formulation.

Theorem 1.2. *Let $F \in k[x_0, x_1, x_2, x_3]$ be a nonzero homogeneous cubic form over a field k. Then F represents zero nontrivially over k if and only if it represents zero nontrivially over a finite field extension L/k of degree coprime to* 3.

By results of Segre, Manin [Man86] and Kollár [Kol02], smooth cubic surfaces with a rational point are unirational. We therefore obtain the following corollary.

Corollary 1.3. *Let S be a smooth cubic surface over a field k. Then S is unirational if and only if it contains a zero-cycle of degree* 1.

In view of [Cor76a, Ma21, Voi26], the key new ingredient in the proof of Theorem 1.2 is Theorem 1.1 above. We give two independent proofs of the latter statement. The first proof, presented in Section 3, uses a reduction of Cassels (see Proposition 2.4 below) to pass from a point of degree 4 to an effective zero-cycle of degree 5. Using work of Buchsbaum–Eisenbud [BE77], Beauville [Beau00] and Tanturri [Tan14], we show that the surface either has a rational point or admits a linear 6×6 Pfaffian presentation over the ground field. The following result then completes the argument.

Theorem 1.4. *Let k be a field of characteristic different from* 2. *If a nonzero cubic form in four variables has a linear 6×6 Pfaffian presentation over k, then its projective zero locus has a k-rational point.*

The second proof of Theorem 1.1, given in Section 4, works directly with the given closed point P of degree 4 on S. We construct a suitable tangent direction at P, giving rise to a nonreduced subscheme Z of S of length 8 that is contained in three linearly independent quadrics Q_1, Q_2, Q_3. If these quadrics have zero-dimensional base locus, then the cubic form F that cuts out S can be written as $F = H_1Q_1 + H_2Q_2 + H_3Q_3$ for linear forms H_i (see Lemma 4.2), and any k-rational point of $H_1 = H_2 = H_3 = 0$ lies on S. If the base locus of the linear series of quadrics through Z is not zero-dimensional, then we show that either S has a rational point or the base locus is a twisted cubic Γ. If Γ lies in S, then S has a rational point because $\Gamma \cong \mathbb{P}^1_k$ (see Lemma 2.3). Otherwise, $\Gamma \cap S$ has degree 9 and contains the subscheme Z of length 8, so the residual intersection has degree 1 and yields a rational point.

The results in this paper have been obtained by or with substantial help from currently available LLMs; see also the AI disclosure section below for further details.

2. Preliminaries and degree reductions

Throughout this section, the field k is perfect. The degree of a finite k-scheme is its length over k. We use the associated effective zero-cycle when taking residual intersections.

The secant and plane-curve descent arguments below are classical; see [Cor76a, Propositions 2.2 and 2.3] for quadratic descent and Poincaré's descent on cubic curves.

Lemma 2.1. *Let $X \subset \mathbb{P}^3_k$ be the zero locus of a nonzero cubic form. If X has an effective zero-cycle of degree one or two, then $X(k) \neq \varnothing$.*

Proof. It suffices to consider a closed point P of degree two. Its two conjugates span a k-line ℓ. If $\ell \subset X$, it supplies a rational point. Otherwise $[\ell \cap X] - [P]$ is an effective zero-cycle of degree one. □

Lemma 2.2. *Let P be a closed point of degree $d \in \{4, 5\}$ on a smooth cubic surface S/k.*

(1) If P is contained in a k-plane, then $S(k) \neq \varnothing$.

(2) *If P is not contained in a k-plane, its geometric points are in general linear position: any four of them span $\mathbb{P}^3$.*

Proof. For (1), let Π contain P. Since $h^0(\Pi, \mathcal{O}_\Pi(2)) = 6 > d$, there is a conic $Q \subset \Pi$ containing P. Write $C = S \cap \Pi$. If C and Q have no common component, then $[C \cap Q] - [P]$ is effective of degree $6 - d \leq 2$, and Lemma 2.1 applies. Otherwise the greatest common divisor of their equations over k has degree one or two. In the first case it defines a k-line on S; in the second, the residual factor of the cubic defines such a line. Part (2) is immediate. □

Lemma 2.3. *Let $\Gamma \subset \mathbb{P}^3_k$ be a smooth projective curve over a field k, whose base change to $\bar{k}$ is a twisted cubic. Then $\Gamma \cong \mathbb{P}^1_k$ and $\Gamma \subset \mathbb{P}^3_k$ is a twisted cubic.*

Proof. Since $\Gamma_{\bar{k}}$ is a twisted cubic, $\mathcal{O}(1)|_\Gamma$ is a line bundle of degree three. Hence, $\omega_\Gamma \otimes \mathcal{O}(1)$ is a line bundle of degree one and so Γ is rational. Under the identification $\Gamma \cong \mathbb{P}^1_k$, the embedding $\Gamma \subset \mathbb{P}^3_k$ must then be given by the full linear series $|\mathcal{O}_{\mathbb{P}^1_k}(3)|$, hence Γ is a twisted cubic. □

The next proposition follows from the $r = 1$ case of Cassels's descent [Cor76a, Proposition 3.1, pp. 274–275]. We give the twisted-cubic construction explicitly.

Proposition 2.4 (Cassels)**.** *A smooth cubic surface S/k has an effective zero-cycle of degree four if and only if it has an effective zero-cycle of degree five. More precisely, a closed point of either degree yields either a rational point or a closed point of the other degree.*

Proof. If $S(k) \neq \varnothing$, multiples of a rational point give both cycles. Otherwise Lemma 2.1 excludes points of degrees one and two; hence an effective zero-cycle of degree $d \in \{4, 5\}$ must be a single closed point P. By Lemma 2.2, its geometric points are in general linear position.

We claim that P lies on a twisted cubic over k. To see this, write $E = \kappa(P) = k(\theta)$ for a primitive element θ and let

$$\nu_3 : \mathbb{P}^1_k \longrightarrow \mathbb{P}^3_k, \qquad [s : t] \longmapsto [s^3 : s^2t : st^2 : t^3].$$

For $d = 4$, choose homogeneous coordinates $P = [a_0 : a_1 : a_2 : a_3]$ with $a_i \in E$. Noncoplanarity means that the a_i form a k-basis of E. Comparing this basis with $1, \theta, \theta^2, \theta^3$ gives a matrix in $\mathrm{GL}_4(k)$ taking $[1 : \theta : \theta^2 : \theta^3]$ to P. Its image of $\nu_3(\mathbb{P}^1)$ is the required curve.

For $d = 5$, let $\theta_1, \ldots, \theta_5$ be the conjugates of θ in a Galois splitting field. The five points $[1 : \theta_i : \theta_i^2 : \theta_i^3]$ for $i = 1, \ldots, 5$ form a projective frame, by the Vandermonde determinant. The corresponding conjugates of P also form a projective frame. There is a unique projectivity matching these two labelled frames. Both labellings have the same Galois action, so uniqueness makes the projectivity Galois invariant; it therefore descends to k. Again its image of $\nu_3(\mathbb{P}^1)$ is a twisted cubic Γ containing P.

If $\Gamma \subset S$, the curve supplies a k-point. Otherwise Bézout's theorem gives

$$\deg(S \cap \Gamma) = 9, \qquad [S \cap \Gamma] - [P] \text{ effective of degree } 9 - d. \tag{2.1}$$

Under the standing assumption $S(k) = \varnothing$, this effective cycle is a single closed point, as above. □

Remark 2.5. When $\mathrm{char}(k) = 0$, the effectivity argument in Voisin's proof of [Voi26, Proposition 4.3(b)] gives another proof.

3. The degree-five argument: Pfaffian presentations

In Section 3.1 we work over a field of characteristic different from two; then in Section 3.2 we restrict ourselves to characteristic zero. Our goal is to prove Theorem 1.4 stated in the introduction and show that it implies the following theorem.

Theorem 3.1. *A smooth cubic surface over a field of characteristic zero with an effective zero-cycle of degree five has a rational point.*

3.1. **Pfaffian cubics have rational points.** Let $V = k^6$ and put $\mathcal{T} = (\bigwedge^2 V^*)^3 \simeq \mathbb{A}^{45}_k$. For an alternating form $A \in \bigwedge^2 V^*$, a subspace $W \subset V$ is isotropic if $A|_{W\times W} = 0$.

Consider the scheme

$$\mathcal{I} = \{(W, A_0, A_1, A_2) : A_i|_{W\times W} = 0 \ (0 \le i \le 2)\} \subset \mathrm{Gr}(3, V) \times \mathcal{T}$$

which parametrizes three-spaces $W \subset V$ that are isotropic for each entry of $(A_0, A_1, A_2) \in \mathcal{T}$. The following lemma is an ordered-triple version of Han's incidence calculation [Han14, Proposition 2.4] (who credits Sasha Kuznetsov).

Lemma 3.2. *There is a nonempty open subset $\mathcal{T}^\circ \subset \mathcal{T}$, defined over k, such that the natural map $\mathcal{I} \times_{\mathcal{T}} \mathcal{T}^\circ \to \mathcal{T}^\circ$ is finite étale of degree two.*

Proof. The restriction map $\bigwedge^2 V^* \to \bigwedge^2 W^*$ is surjective. Thus $\mathcal{I}$ is the total space of a rank-36 vector bundle over $\mathrm{Gr}(3, 6)$, and is smooth of dimension 45. The projection $\pi : \mathcal{I} \to \mathcal{T}$ is proper.

Write $V = V_1 \oplus V_2 \oplus V_3$, with $\dim V_i = 2$, and set

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Consider the triple of alternating 6×6 matrices

$$B_0 = \mathrm{diag}(J, J, J), \qquad B_1 = \mathrm{diag}(J, 2J, 3J), \qquad B_2 = \begin{pmatrix} 0 & I_2 & I_2 \\ -I_2 & 0 & D \\ -I_2 & -D & 0 \end{pmatrix}. \tag{3.1}$$

A B_0-isotropic three-space W is Lagrangian with respect to the standard symplectic form on $V = k^6$. Put $T = B_0^{-1}B_1 = \mathrm{diag}(I_2, 2I_2, 3I_2)$. If W is also B_1-isotropic, then

$$B_0(Tw, w') = B_1(w, w') = 0 \quad (w, w' \in W), \qquad T(W) \subset W^{\perp_{B_0}} = W.$$

Thus W is T-invariant and splits into its intersections with the three eigenspaces. Isotropy bounds the dimension of each summand by one, so $W = \ell_1 \oplus \ell_2 \oplus \ell_3$ with $\ell_i \subset V_i$ a line. This description holds scheme-theoretically. The common isotropic locus for B_0, B_1 is therefore $(\mathbb{P}^1)^3$. Writing $\ell_i = [a_i : b_i]$, the remaining equations are

$$a_1a_2 + b_1b_2 = 0, \qquad a_1a_3 + b_1b_3 = 0, \qquad a_2a_3 + 2b_2b_3 = 0.$$

No a_i vanishes at a geometric solution: if $a_1 = 0$ then $b_2 = b_3 = 0$, contradicting the third equation; the other cases are similar. Set $u_i = b_i/a_i$. The fiber is

$$u_2 = u_3 = -u_1^{-1}, \qquad u_1^2 + 2 = 0, \tag{3.2}$$

and hence is finite and geometrically reduced of degree two.

At these two geometric points the differential of π is an isomorphism, since source and target are smooth of the same dimension and the fiber tangent spaces vanish. Thus π is *étale* there. Properness allows us to remove the image of its non-*étale* locus and shrink around (B_0, B_1, B_2); the resulting morphism is proper and quasi-finite, hence finite *étale*, of degree two. □

Proof of Theorem 1.4. Let k be a field of characteristic different from 2 and let $S \subset \mathbb{P}^3_k$ be a cubic surface, cut out by $F = 0$. Assume that

$$F(x) = \mathrm{Pf}\left(\sum_{i=0}^{3} x_iA_i\right),$$

where the A_i are alternating matrices over k.

We first deal with the case where $(A_0, A_1, A_2) \in \mathcal{T}^\circ(k)$. Then Lemma 3.2 gives an extension L/k of degree at most two and a three-space $W \subset L^6$ isotropic for A_0, A_1, A_2.

The alternating form $A_3|_W$ has a nonzero radical vector $v \in W$, since $\dim W = 3$. Consequently all four covectors

$$A_0v, \ A_1v, \ A_2v, \ A_3v$$

belong to the three-dimensional subspace $\operatorname{Ann}(W) \subset (L^6)^*$ of linear forms $\lambda\colon L^6 \to L$ that vanish on W. Choose a nontrivial relation $\sum b_i(A_i v) = 0$ with $b_i \in L$. The alternating matrix $\sum b_i A_i$ is degenerate, so

$$F(b_0, b_1, b_2, b_3)^2 = \det\left(\sum_{i=0}^{3} b_i A_i\right) = 0.$$

This gives an L-point of $F = 0$. Its image in S has degree at most two, and Lemma 2.1 gives a k-point.

For an arbitrary triple $(A_0, A_1, A_2) \in \mathcal{T}(k)$, use the fixed matrices (3.1) and put

$$A_i(t) = (1-t)A_i + tB_i \quad (0 \le i \le 2), \qquad A_3(t) = A_3, \qquad F_t(x) = \operatorname{Pf}\left(\sum_{i=0}^{3} x_i A_i(t)\right).$$

The generic triple belongs to $\mathcal{T}^\circ$ because its value at $t = 1$ does; moreover $F_t \neq 0$ over $k(t)$ since $F_0 = F \neq 0$. The preceding argument over $k(t)$ gives a $k(t)$-point of $F_t = 0$. Scale its homogeneous coordinates into the discrete valuation ring $k[t]_{(t)}$, with at least one a unit. Reduction modulo t then gives a k-point of $F = 0$. □

3.2. **Five points and the characteristic-zero theorem.**

Lemma 3.3 (Buchsbaum–Eisenbud, Beauville, Tanturri)**.** *Let $Z \subset \mathbb{P}^3_k$ be a reduced subscheme of degree five whose geometric points are in general linear position. Every cubic form vanishing on Z has a linear 6×6 Pfaffian presentation over k.*

Proof. Put $R = k[x_0, x_1, x_2, x_3]$. The ideal of five points in general linear position is arithmetically Gorenstein, with resolution

$$0 \longrightarrow R(-5) \longrightarrow R(-3)^5 \longrightarrow R(-2)^5 \longrightarrow R \longrightarrow R/I_Z \longrightarrow 0. \tag{3.3}$$

For completeness, this standard five-point calculation can be checked after extending to $\overline{k}$ and moving the points to $[e_0], [e_1], [e_2], [e_3], [1:1:1:1]$. Their ideal is generated by the five independent differences among the six products $x_i x_j$ with $i < j$. In the quotient all mixed monomials of a fixed degree agree, and the Hilbert function is $1, 4, 5, 5, \ldots$ (equal to 5 for all degrees $d \ge 2$). Reducing by the nonzerodivisor $\ell = x_0 + x_1 + x_2 + x_3$, and writing q for the class represented by $x_i x_j$ with $i \neq j$ gives

$$x_i^2 = -3q, \qquad x_i x_j = q \ (i \neq j), \qquad x_i q = 0.$$

On the basis x_0, x_1, x_2 of degree one, multiplication into degree two is represented by

$$\begin{pmatrix} -3 & 1 & 1 \\ 1 & -3 & 1 \\ 1 & 1 & -3 \end{pmatrix}, \qquad \det = -16.$$

Thus the socle is one-dimensional in degree two. The ring is Gorenstein, and its five quadratic generators and self-duality give (3.3). Gorensteinness and the graded Betti numbers descend by faithful flatness.

Apply the Buchsbaum–Eisenbud structure theorem [BE77] over k. The middle map of (3.3) is represented by a 5×5 alternating matrix T of linear forms, and I_Z is generated by

$$q_i = (-1)^{i+1} \operatorname{Pf}(T_{\widehat{i}}) \qquad (1 \le i \le 5),$$

where $T_{\widehat{i}}$ are the principal submatrices of T. If F is a cubic vanishing on Z, write $F = \sum_i \ell_i q_i$ with linear forms ℓ_i over k. Expanding along the last column gives

$$F = \operatorname{Pf}\begin{pmatrix} T & \ell \\ -\ell^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \ell = (\ell_1, \ldots, \ell_5)^{\mathsf{T}}. \tag{3.4}$$

This is the construction in [Beau00, Remark 7.3(b)] and [Tan14, Section 2]. □

Proof of Theorem 3.1. If the cycle has a point of degree one or two in its support, apply Lemma 2.1. Otherwise it is a single closed point P of degree five. The coplanar case is Lemma 2.2(1). In the remaining case Lemma 2.2(2) gives general linear position, Lemma 3.3 supplies a Pfaffian presentation over k, and Theorem 1.4 gives a rational point. □

4. The degree-four argument: nets of quadrics

Combining Theorem 3.1 and Proposition 2.4 proves Theorem 1.1 stated in the introduction, which assumes $\mathrm{char}(k) = 0$. In this section we give a second proof, which naturally works over any perfect field and yields the following theorem.

Theorem 4.1. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field. If S has a closed point P of degree four, then $S(k) \neq \varnothing$.*

The purpose of this section is to give a direct proof of Theorem 4.1 which works directly with the given degree 4 point.

4.1. **A rational-point criterion from three quadrics.** Let $S = \{F = 0\} \subset \mathbb{P}^3_k$ be a smooth cubic surface over a field k. If S has a k-rational point P, then we may after a change of coordinates assume that $P = [1 : 0 : 0 : 0]$, in which case $F = x_1Q_1 + x_2Q_2 + x_3Q_3$ for quadratic forms Q_i. Conversely, if

$$F = H_1Q_1 + H_2Q_2 + H_3Q_3 \in k[x_0, x_1, x_2, x_3] \tag{4.1}$$

for linear and quadratic forms $H_i \in k[x_0, x_1, x_2, x_3]$ and $Q_i \in k[x_0, x_1, x_2, x_3]$, respectively, then S contains the linear space $\{H_1 = H_2 = H_3 = 0\} \subset \mathbb{P}^3_k$ and hence a rational point.

Our strategy is to find a decomposition as in (4.1).

Lemma 4.2. *Let $S = \{F = 0\} \subset \mathbb{P}^3_k$ be a smooth cubic surface over a field k. Let $Z \subset S$ be a zero-dimensional subscheme of length 8 and assume that there are three quadratic forms $Q_1, Q_2, Q_3 \in k[x_0, x_1, x_2, x_3]$ such that $W := \{Q_1 = Q_2 = Q_3 = 0\} \subset \mathbb{P}^3_k$ has dimension zero and contains Z. Then there are linear forms $H_i \in k[x_0, x_1, x_2, x_3]$ such that (4.1) holds.*

Proof. Since W has dimension zero and contains Z, we must have $W = Z$, because W has length $2^3 = 8$. Hence, $\mathcal{I}_Z = (Q_1, Q_2, Q_3)$ and Q_1, Q_2, Q_3 form a regular sequence. The Koszul complex therefore yields the exact sequence

$$0 \longrightarrow \mathcal{O}(-6) \longrightarrow \mathcal{O}(-4)^{\oplus 3} \longrightarrow \mathcal{O}(-2)^{\oplus 3} \longrightarrow \mathcal{I}_Z \longrightarrow 0 \tag{4.2}$$

on $\mathbb{P}^3$. Twisting with $\mathcal{O}(3)$, this yields the exact sequence

$$0 \longrightarrow \mathcal{O}(-3) \longrightarrow \mathcal{O}(-1)^{\oplus 3} \longrightarrow \mathcal{O}(1)^{\oplus 3} \longrightarrow \mathcal{I}_Z(3) \longrightarrow 0.$$

Since $H^{i+1}(\mathbb{P}^3, \mathcal{O}(-3)) = H^i(\mathbb{P}^3, \mathcal{O}(-1)) = 0$ for $i = 0, 1$, we conclude that

$$H^0(\mathbb{P}^3, \mathcal{O}(1)^{\oplus 3}) \xrightarrow{\cong} H^0(\mathbb{P}^3, \mathcal{I}_Z(3)), \qquad (H_1, H_2, H_3) \mapsto \sum_{i=1}^{3} H_iQ_i$$

is an isomorphism. Finally, since $Z \subset S$, the cubic form F yields a global section of $\mathcal{I}_Z(3)$ and we obtain a decomposition as in (4.1). □

Remark 4.3. Twisting (4.2) with $\mathcal{O}(2)$ and taking cohomology, one obtains similarly that $H^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \cong H^0(\mathbb{P}^3, \mathcal{O}^{\oplus 3})$ has dimension 3 in the setting of Lemma 4.2.

Proposition 4.4. *Let $S = \{F = 0\}$ be a smooth cubic surface over a field k. Let $Z \subset S$ be a zero-dimensional subscheme of length 8. Assume that the base locus of the linear series $|\mathcal{I}_Z(2)|$ on $\mathbb{P}^3$ is zero-dimensional. Then S admits a rational point.*

Proof. Since the base locus of $|\mathcal{I}_Z(2)|$ is zero-dimensional, we can find, after base change to an infinite field, three linearly independent quadrics which have zero-dimensional intersection. Hence, $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) = h^0(\mathbb{P}^3_{\bar{k}}, \mathcal{I}_{Z_{\bar{k}}}(2)) = 3$ by Remark 4.3. It follows that there are three quadratic forms Q_1, Q_2, Q_3 which satisfy the assumptions in Lemma 4.2. Hence, F has a decomposition as in (4.1) and so S has a rational point. □

A necessary condition for the base locus of $|\mathcal{I}_Z(2)|$ to be zero-dimensional is $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We will study this condition and the resulting base loci in the next subsection; since base loci are compatible with the extension of the ground field, it will be enough to treat the geometric case where the ground field is algebraically closed.

4.2. **Quadrics through four tangent double points: a geometric analysis.** In this subsection we work over an algebraically closed field k. Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface and let $p_0, \dots, p_3 \in S$ be closed points that do not lie on a plane. Up to a coordinate change, we may assume that $p_i = [e_i]$ are the four coordinate vertices in $\mathbb{P}^3_k$, i.e. the points represented by the standard basis in k^4. Let $Z = \sqcup_{i=0}^3 Z_i$ be a length 8 subscheme of S, where Z_i is of length 2 and supported at the point p_i.

We may represent Z_i by $[e_i + \epsilon v_i]$, where $\epsilon^2 = 0$ and $v_i = (u_{i0}, \dots, u_{i3})$ is a nontrivial tangent direction at p_i with $u_{ii} = 0$. That is, the line $\{[se_i + tv_i] \mid [s : t] \in \mathbb{P}^1\} \subset \mathbb{P}^3$ is contained in the projective tangent space of S at p_i. In the affine chart $x_i = 1$, with coordinates $y_j = x_j/x_i$ for $j \neq i$, the tangent direction v_i may equivalently be described by

$$v_i = \sum_{j \neq i} u_{ij} \frac{\partial}{\partial y_j} \in T_{p_i} S. \tag{4.3}$$

Lemma 4.5. *In the above notation, we have $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3$ if and only if there are scalars $\lambda_0, \dots, \lambda_3 \in k$, not all zero, such that*

$$\lambda_i u_{ij} = -\lambda_j u_{ji} \qquad \text{for all } 0 \leq i < j \leq 3.$$

Proof. A quadric through $p_0, \dots, p_3$ has the form

$$Q = \sum_{0 \leq i < j \leq 3} q_{ij} x_i x_j. \tag{4.4}$$

The space of such quadrics has dimension $\binom{4}{2} = 6$.

For convenience, set $q_{ji} = q_{ij}$ whenever $i < j$. Since $Q(p_i) = 0$, the quadric Q vanishes along Z_i if and only if $d_{p_i} Q(v_i) = 0$. Moreover,

$$d_{p_i} Q(v_i) = \sum_{j \neq i} q_{ij} u_{ij}. \tag{4.5}$$

Thus the quadrics vanishing along Z form the kernel of the linear map

$$H^0(\mathbb{P}^3, \mathcal{I}_{Z_{\mathrm{red}}}(2)) \longrightarrow k^4, \qquad Q \mapsto (d_{p_0} Q(v_0), \dots, d_{p_3} Q(v_3))$$

from a six-dimensional vector space to k^4. This kernel has dimension at least 3 if and only if the four linear functionals $Q \mapsto d_{p_i} Q(v_i)$ are linearly dependent. This is equivalent to the existence of scalars λ_i, not all zero, such that

$$0 = \sum_{i=0}^{3} \lambda_i d_{p_i} Q(v_i) = \sum_{i<j} q_{ij} (\lambda_i u_{ij} + \lambda_j u_{ji})$$

for every Q. Since the coefficients q_{ij} are arbitrary, this is equivalent to $\lambda_i u_{ij} = -\lambda_j u_{ji}$ for every $i < j$, as we want. □

The previous lemma gave a necessary and sufficient condition for $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We will concentrate on the case $u_{ij} = -u_{ji}$ for all i, j, which ensures $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$. We further assume $u_{ij} \neq 0$ for $i \neq j$ to rule out degenerate cases (which will be dealt with later in Proposition 4.8 via a different argument). The next lemma characterizes the possible base loci in this situation.

Lemma 4.6. *Assume in the above notation that $u_{ij} = -u_{ji}$ for all i, j and $u_{ij} \neq 0$ for $i \neq j$. Then the base locus B of the linear series $|\mathcal{I}_Z(2)| \subset |\mathcal{O}_{\mathbb{P}^3}(2)|$ is one of the following:*

(1) $B = Z$, or
(2) B is a twisted cubic in $\mathbb{P}^3$.

Proof. Consider the matrix $U := (u_{ij}) \in k^{4\times 4}$ which encodes the tangent directions of Z at the four coordinate vertices. For convenience, we rename the u_{ij} by a, b, c, d, e, f and get

$$U = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix}, \qquad \text{with} \quad abcdef \neq 0.$$

A direct computation using (4.5) shows that the quadratic forms

$$\begin{aligned} Q_1 &:= \frac{d}{a}x_0x_1 - \frac{d}{b}x_0x_2 + x_1x_2, \\ Q_2 &:= \frac{e}{a}x_0x_1 - \frac{e}{c}x_0x_3 + x_1x_3, \\ Q_3 &:= \frac{f}{b}x_0x_2 - \frac{f}{c}x_0x_3 + x_2x_3, \end{aligned}$$

are linearly independent and satisfy $d_{p_i}Q_j(v_i) = 0$ for all $i = 0, 1, 2, 3$ and all $j = 1, 2, 3$. Hence, $Q_1, Q_2, Q_3 \in H^0(\mathbb{P}^3, \mathcal{I}_Z(2))$ span a three-dimensional subspace.

We aim to describe the base locus $B' := \{Q_1 = Q_2 = Q_3 = 0\}$. If this is zero-dimensional, then B must be equal to Z. It thus suffices to show that B' is either zero-dimensional or a twisted cubic. Indeed, if B' is a twisted cubic, then so is B, because zero-dimensional B implies $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) = 3$ by Remark 4.3 (cf. proof of Proposition 4.4).

On the plane $x_0 = 0$, the common zero-set of these quadrics is given by the coordinate vertices p_1, p_2, p_3. It remains to analyze the affine chart $x_0 = 1$ and we write $x_1 = t$. The vanishing $Q_1 = Q_2 = 0$ gives

$$x_2 = \frac{bdt}{a(d - bt)} \qquad \text{and} \qquad x_3 = \frac{cet}{a(e - ct)}. \tag{4.6}$$

One checks that neither denominator can vanish at a point of the base locus B'. Substitution into the third equation gives

$$Q_3 = \frac{t^2\Delta}{a^2(d - bt)(e - ct)} \qquad \text{where} \quad \Delta := abef - acdf + bcde.$$

If $\Delta \neq 0$, the base scheme B' is therefore finite. In fact, the above equation gives a base scheme of length 2 on the chart $x_0 = 1$, which is compatible with the fact that we work away from p_1, p_2, p_3.

Suppose now that $\Delta = 0$. Then the vanishing of Q_3 is automatic, once (4.6) holds. Moreover, $be - cd \neq 0$, since otherwise $\Delta = bcde \neq 0$. We introduce a second variable s, homogenize (4.6) with respect to s, and clear denominators to find that the base locus is contained in the projective curve described by

$$\begin{aligned} x_0 &= as(ds - bt)(es - ct), \\ x_1 &= at(ds - bt)(es - ct), \\ x_2 &= bdst(es - ct), \\ x_3 &= cest(ds - bt), \end{aligned}$$

with $[s : t] \in \mathbb{P}^1$. Note that each x_i is homogeneous of degree 3 in s, t, hence may be viewed as an element of $H^0(\mathbb{P}^1, \mathcal{O}(3))$. We check that the coefficient matrix of $x_0, \ldots, x_3$ in the basis s^3, s^2t, st^2, t^3 has determinant $-a^2b^2c^2d^2e^2(be - cd) \neq 0$. Hence, the above expressions for $x_0, \ldots, x_3$ form a basis of $H^0(\mathbb{P}^1, \mathcal{O}(3))$, hence they parametrize a smooth twisted cubic $\Gamma \subset \mathbb{P}^3$.

To double check our computation, we substitute into Q_1 and Q_2 and get zero, while substitution into Q_3 gives

$$s^2t^2(ds - bt)(es - ct)\Delta = 0.$$

Thus all three quadrics vanish on Γ. The ideal sheaf of a twisted cubic is always generated by three quadrics (see [Eis05, Proposition 6.1]) and so we must have $\mathcal{I}_\Gamma = (Q_1, Q_2, Q_3)$. This concludes the proof of the lemma. □

4.3. **First arithmetic consequences.** We first show that the assumption $u_{ij} = -u_{ji}$ used in Lemma 4.6 is automatic in the arithmetic situation, as follows.

Lemma 4.7. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $Z \subset S$ be a subscheme of length 8 supported at a closed point of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. If $h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3$, then, over $\bar{k}$, and in coordinates such that Z is supported on the coordinate vertices $p_i = [e_i]$, the tangent directions can be chosen so that $u_{ij} = -u_{ji}$ for all i, j in (4.3).*

Proof. Work over $\bar{k}$ and take the four geometric support points as coordinate vertices. Using the row convention for the tangent representatives, Lemma 4.5 shows that

$$R = \left\{ (\lambda_i) \in \bar{k}^4 \;\middle|\; \lambda_i u_{ij} + \lambda_j u_{ji} = 0 \text{ for all } i < j \right\}$$

is nonzero.

Since Z is defined over k, Galois acts semilinearly on R, permuting the coordinates up to nonzero scalar factors. This permutation action is transitive because the support of Z is a single closed point. Thus, if R were contained in one coordinate hyperplane $\{\lambda_i = 0\}$, it would be contained in all four, contradicting $R \neq 0$.

Since $\bar{k}$ is infinite, the four proper linear subspaces $R \cap \{\lambda_i = 0\}$ cannot cover R. Choose $(\lambda_i) \in R$ with every $\lambda_i \neq 0$. Replacing v_i by $\lambda_i v_i$ preserves Z_i and gives

$$\widetilde{u}_{ij} = \lambda_i u_{ij} = -\lambda_j u_{ji} = -\widetilde{u}_{ji},$$

as required. □

Lemma 4.6 describes the base locus in the case where $u_{ij} = -u_{ji}$ is nonzero for $i \neq j$. The missing case will be handled through the following ad hoc argument.

Proposition 4.8. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $P \in S$ be a closed point of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. Let $Z \subset S$ be a subscheme of length 8 with $Z_{\mathrm{red}} = P$. Suppose that, after choosing coordinates in which the four geometric points of P are the coordinate vertices $p_i = [e_i]$, we can write*

$$Z_{\bar{k}} = \coprod_{i=0}^{3} Z_i, \qquad Z_i = \{[e_i + \epsilon v_i] : \epsilon^2 = 0\}, \qquad v_i = \sum_{j=0}^{3} u_{ij} e_j,$$

where every v_i is nonzero and $U = (u_{ij})$ is skew-symmetric with $u_{ii} = 0$ for all i.

If $u_{ij} = 0$ for some $i \neq j$, then $S(k) \neq \varnothing$.

Proof. Consider the graph Γ on the set of vertices $\{0, 1, 2, 3\}$ whose edges are the unordered pairs $\{i, j\}$ with $u_{ij} \neq 0$. This graph carries a natural action by the Galois group of k. The Galois action on the four vertices is transitive, so this graph is regular, i.e. there is a positive integer n such that its valence at each vertex is n. Since every v_i is nonzero, while some u_{ij} with $i \neq j$ are zero, we find $n \in \{1, 2\}$.

Suppose first that $n = 1$. After relabeling, the two edges are $\{0, 1\}$ and $\{2, 3\}$. For $i \neq j$, write $L_{ij} = \langle p_i, p_j \rangle$ for the line spanned by p_i and p_j. Then the subscheme $Z_0 \sqcup Z_1$ is contained in L_{01}, so the restriction of the cubic equation of S to L_{01} vanishes on a subscheme of length 4. Hence, $L_{01} \subset S_{\bar{k}}$. A similar argument shows $L_{23} \subset S_{\bar{k}}$.

The lines L_{01} and L_{23} have no point in common and their union is Galois invariant. In fact, the Galois action interchanges the two lines and so their union is defined over k and there is a quadratic extension K/k over which L_{01} and L_{23} are defined. Cubic surfaces that contain such skew lines over a quadratic extension are rational, hence contain in particular a rational point. To construct such a point explicitly, choose $a \in L_{01}(K)$ and a generator

$\sigma \in \mathrm{Gal}(K/k)$. Then the line $M = \langle a, \sigma(a)\rangle$ is defined over k. If $M \subset S$, it supplies a k-rational point. Otherwise the residual intersection point of $M \cap S$ provides such a point.

Suppose now that $n = 2$. Then Γ is the four-cycle graph C_4. After relabeling, we may assume that

$$u_{01} = u_{23} = 0, \qquad b = u_{02}, \quad c = u_{03}, \quad d = u_{12}, \quad e = u_{13}, \qquad bcde \neq 0.$$

The three quadrics

$$Q_1 = x_0x_1, \qquad Q_2 = x_2x_3, \qquad Q_3 = \frac{x_0x_2}{b} - \frac{x_0x_3}{c} - \frac{x_1x_2}{d} + \frac{x_1x_3}{e}$$

vanish on $Z_{\bar{k}}$. Indeed, they vanish at the four vertices, and direct substitution using (4.5) gives $d_{p_i}Q_j(v_i) = 0$ for every i, j. Moreover,

$$\{Q_1 = Q_2 = 0\} = L_{02} \cup L_{03} \cup L_{12} \cup L_{13}.$$

The form Q_3 restricts nontrivially on each of these lines. Thus $\{Q_1 = Q_2 = Q_3 = 0\}$ is zero-dimensional. In particular, the base locus of $|\mathcal{I}_{Z_{\bar{k}}}(2)|$ is zero-dimensional.

Formation of the base locus commutes with extension of the ground field, so the base locus of $|\mathcal{I}_Z(2)|$ over k is also zero-dimensional and Proposition 4.4 gives $S(k) \neq \varnothing$. □

4.4. **Construction of the length-8 subscheme.** We now construct a subscheme $Z \subset S$ of length 8 over the ground field by choosing a suitable tangent direction at a closed point of degree 4.

Proposition 4.9. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over a perfect field k, and let $P \in S$ be a closed point of degree 4 whose geometric points span $\mathbb{P}^3_{\bar{k}}$. Then there exists a subscheme $Z \subset S$, defined over k and supported at P, such that $Z_{\bar{k}}$ is a disjoint union of four subschemes of length 2 and*

$$h^0(\mathbb{P}^3_k, \mathcal{I}_Z(2)) \geq 3.$$

Proof. Put $L = \kappa(P)$, and let $\mathcal{J}_P \subset \mathcal{O}_S$ be the ideal sheaf of P in S. Since S is smooth and L/k is separable, the restriction of the cotangent bundle $\Omega^1_S := \Omega^1_{S/k}$ to P

$$\Omega^1_S|_P := \Omega^1_{S/k} \otimes_{\mathcal{O}_S} L \cong \mathcal{J}_P/\mathcal{J}_P^2$$

has dimension 2 over L. The dual of this space is the tangent space $T_S|_P$ at P.

The choice of a subscheme $Z \subset S$ of length 8 which is supported at the point P and whose base change to $\bar{k}$ splits into 4 subschemes of length 2 is equivalent to the choice of a line in the L-vector space $T_S|_P$. Dually, it corresponds to a one-dimensional quotient of $\Omega^1_S|_P$.

Set

$$E := \Omega^1_S(2)|_P \qquad \text{and} \qquad V := H^0(\mathbb{P}^3_k, \mathcal{I}_P(2)),$$

where $\mathcal{I}_P$ is the ideal sheaf of P in $\mathbb{P}^3_k$. The four geometric points of P impose independent conditions on quadrics, since they span $\mathbb{P}^3_{\bar{k}}$. Thus $\dim_k V = 6$. Moreover, $\dim_L E = 2$ and $\dim_k E = 8$. Restricting a quadric which vanishes at P to S and reducing modulo $\mathcal{J}_P^2$ defines a canonical k-linear map

$$\varphi\colon V \longrightarrow H^0(P, (\mathcal{J}_P/\mathcal{J}_P^2) \otimes_L \mathcal{O}_S(2)|_P) = \Omega^1_S(2)|_P,$$

where we use that $\Omega^1_S(2)|_P = (\mathcal{J}_P/\mathcal{J}_P^2) \otimes_L \mathcal{O}_S(2)|_P$.

Let us first express the desired property in terms of this map. Given any L-linear quotient $q\colon E \twoheadrightarrow N$ with $\dim_L N = 1$, tensoring with $\mathcal{O}_S(-2)|_P$ yields a one-dimensional quotient of $\Omega^1_S|_P$, hence a length 8 subscheme Z_q supported at P as above. A quadric $Q \in V$ vanishes on Z_q precisely when $q(\varphi(Q)) = 0$. Therefore

$$H^0(\mathbb{P}^3_k, \mathcal{I}_{Z_q}(2)) = \ker\big(q \circ \varphi\colon V \longrightarrow N\big). \tag{4.7}$$

Since $\dim_k V = 6$ and $\dim_k N = 4$, the desired inequality $h^0(\mathbb{P}^3_k, \mathcal{I}_{Z_q}(2)) \geq 3$ is equivalent to $\mathrm{rank}_k(q \circ \varphi) \leq 3$. Equivalently, there must exist a nonzero k-linear functional $\mu\colon N \to k$ such

that $\mu \circ q \circ \varphi = 0$. Pulling this functional back along q gives a nonzero k-linear functional $\lambda := \mu \circ q \colon E \to k$ satisfying

$$\lambda(\varphi(Q)) = 0 \qquad \text{for every } Q \in V. \tag{4.8}$$

We will first choose such a functional λ and then recover an L-linear one-dimensional quotient of E through which it factors.

Consider the k-vector space

$$K = \ker\left(\varphi^\vee \colon \operatorname{Hom}_k(E,k) \longrightarrow \operatorname{Hom}_k(V,k)\right), \qquad \varphi^\vee(\lambda) = \lambda \circ \varphi.$$

The source has dimension 8 and the target dimension 6, so $\dim_k K \geq 2$. We may therefore choose a nontrivial element $\lambda \in K$. This gives a k-linear functional on E which satisfies (4.8).

It remains to show that the k-linear map λ factors through some one-dimensional L-linear quotient $q \colon E \to N \cong L$ of E. To this end, we use the k-linear isomorphism

$$\begin{aligned} \Theta : \operatorname{Hom}_L(E,L) &\xrightarrow{\sim} \operatorname{Hom}_k(E,k), \\ f &\mapsto \operatorname{Tr}_{L/k} \circ f, \end{aligned} \tag{4.9}$$

provided by the trace map. (To see that Θ is an isomorphism of k-vector spaces, it suffices to note that it is k-linear and injective, while source and target have the same finite dimension over k. To see injectivity, note that any nonzero L-linear map $f \colon E \to L$ is surjective and $\operatorname{Tr}_{L/k}$ is surjective because L/k is separable, hence $\operatorname{Tr}_{L/k} \circ f$ is surjective and therefore nonzero as long as f is nonzero.)

It follows that there is a unique nonzero L-linear map $f \colon E \to L$ such that $\lambda = \operatorname{Tr}_{L/k} \circ f$. Since λ is nonzero, f must be surjective, and λ factors through it; the role of $\mu \colon N \cong L \to k$ above is then taken by the trace map $\operatorname{Tr}_{L/k}$. As explained above, the length 8 subscheme that is associated to the one-dimensional quotient f has then the properties asserted in the proposition. This concludes the proof. □

4.5. **Proof of Theorem 4.1.**

Proof of Theorem 4.1. By Lemma 2.2, we may assume that the geometric points associated to P span $\mathbb{P}^3_{\bar{k}}$. By Proposition 4.9, there is a closed subscheme $Z \subset S$ of length 8 supported on P such that $Z_{\bar{k}}$ is a disjoint union of four subschemes of length 2 and such that $h^0(\mathbb{P}^3, \mathcal{I}_Z(2)) \geq 3$.

We pass to the algebraic closure $\bar{k}$ of k and choose coordinates on $\mathbb{P}^3_{\bar{k}}$ such that the four geometric points $p_0, \dots, p_3$ that correspond to P agree with the four coordinate vertices: $p_i = [e_i]$. We may then write $Z_{\bar{k}} = \sqcup_{i=0}^3 Z_i$, where Z_i is the length 2 subscheme supported at p_i that is given by $[e_i + \epsilon v_i]$ with $\epsilon^2 = 0$ and $v_i = (u_{i0}, \dots, u_{i3})$ with $u_{ii} = 0$. By Lemma 4.7, we can arrange that $u_{ij} = -u_{ji}$. By Proposition 4.8 we may further assume that u_{ij} is nonzero for all $i \neq j$. It then follows from Lemma 4.6 that the base locus of $|\mathcal{I}_{Z_{\bar{k}}}(2)|$ is either finite or a twisted cubic. It follows that the base locus of the linear series $|\mathcal{I}_Z(2)|$ on $\mathbb{P}^3_k$ over the ground field k is either finite or a smooth curve Γ whose base change to $\bar{k}$ is a twisted cubic.

If the base locus of $|\mathcal{I}_Z(2)|$ is zero-dimensional, then S has a rational point by Proposition 4.4. Otherwise, the base locus is a smooth geometrically rational curve $\Gamma \subset \mathbb{P}^3_k$ such that $\Gamma_{\bar{k}}$ is a twisted cubic. By Lemma 2.3, $\Gamma \cong \mathbb{P}^1_k$ is a twisted cubic, hence has a rational point. If $\Gamma \subset S$, then S has a rational point because $\Gamma \cong \mathbb{P}^1_k$. Otherwise, $\Gamma \cap S$ is a zero-dimensional subscheme of length 9 which contains the length 8 subscheme Z; the residual intersection point then yields a rational point. This concludes the proof of the theorem. □

5. From characteristic zero to positive characteristic

We begin with the following.

Theorem 5.1. *A smooth cubic surface over a field of characteristic zero with a zero-cycle of degree one has a rational point.*

Proof. Voisin's theorem [Voi26, Theorem 1.5], which improved Coray's reduction [Cor76a, Theorem 7.1], gives a point of degree one or four. We may then either apply Theorem 4.1, or use Proposition 2.4 to produce an effective zero-cycle of degree five and apply Theorem 3.1. □

In the remainder of this section, we reduce the case of arbitrary base fields (including imperfect ones) to the case of characteristic zero. The required lifting statement is the smooth-surface instance of Ma's method [Ma21, Section 2.2, Lemmas 2.4–2.5]; we include its proof.

Lemma 5.2. *Let k have characteristic $p > 0$, let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface, and let $P \subset S$ be a closed point of degree d. There exist a complete discrete valuation ring R of characteristic zero with residue field k, a smooth cubic surface $\mathscr{S} \subset \mathbb{P}^3_R$ with special fiber S, and a finite flat subscheme $\mathscr{Z} \subset \mathscr{S}$ of degree d with special fiber P.*

Proof. Choose a Cohen ring R with uniformizer p and residue field exactly k [Stacks, Tag 0328]. Lift the coefficients of a defining cubic of S to obtain $\mathscr{S} \subset \mathbb{P}^3_R$. This hypersurface is flat over R and smooth along its special fiber. Its nonsmooth locus is closed and proper over R. If nonempty, its image would contain the closed point of $\operatorname{Spec} R$, a contradiction. Thus $\mathscr{S}/R$ is smooth.

The closed immersion $P \hookrightarrow S$ is a local complete intersection: the maximal ideal of the regular local ring $\mathcal{O}_{S,P}$ is generated by a regular sequence. By the infinitesimal lifting theorem for finite local complete intersections, the relative Hilbert functor is formally smooth at $[P]$ [Stacks, Tag 06D9]. Since the relative Hilbert scheme $\mathrm{Hilb}^d(\mathscr{S}/R)$ is locally of finite presentation, it is smooth over R near $[P]$; see also [Fog68, Theorem 2.9] and [Ma21, Lemma 2.5]. Since R is complete, hence henselian, its k-point $[P]$ lifts to an R-point. The universal subscheme gives $\mathscr{Z}$. □

We are now ready to prove the case of smooth cubic surfaces.

Theorem 5.3. *Let $S \subset \mathbb{P}^3_k$ be a smooth cubic surface over an arbitrary field. If S has a zero-cycle of degree one, then $S(k) \neq \varnothing$.*

Proof. By Theorem 5.1, only positive characteristic remains. Choose a closed point P of degree d prime to three from the support of a zero-cycle of degree one. Apply Lemma 5.2, and put $K = \operatorname{Frac}(R)$. The finite scheme $\mathscr{Z}_K$ has degree d, so its fundamental cycle, together with the degree-three intersection class with a line, gives a zero-cycle of degree one on $\mathscr{S}_K$.

Theorem 5.1 gives a K-point of $\mathscr{S}_K$. Properness extends it to an R-point of $\mathscr{S}$, whose reduction is a k-point of S. □

6. Removing the smoothness hypothesis

In this section we finish the proof of the following theorem, which is equivalent to Theorem 1.2.

Theorem 6.1. *Let $X = \{F = 0\} \subset \mathbb{P}^3_k$ be a cubic surface over a field k. Then X has a k-rational point if and only if it has a closed point of degree coprime to 3, or, equivalently, a zero-cycle of degree one.*

For singular cubic surfaces over perfect fields, the conclusion of Theorem 6.1 is due to Coray [Cor76a, Proposition 8.2]; see also [Cor76b, Corollary 1] for a second proof. We give a reduction to the smooth case over a Laurent-series field that applies over arbitrary fields, without integrality or reducedness assumptions.

Let $X = \{F = 0\} \subset \mathbb{P}^3_k$ for a nonzero homogeneous cubic F, with no assumption on k or on the singularities of X. If F is reducible over k, it has a linear factor ℓ, so X contains the k-plane $V(\ell)$ and $X(k) \neq \varnothing$. Write X_{sm} for the smooth locus over k. We first handle the possibility that a zero-cycle of degree one is supported entirely in the singular locus.

Lemma 6.2. *Let L/k be a finite extension with $3 \nmid [L : k]$. If $X(L) \neq \varnothing$, then either $X(k) \neq \varnothing$ or $X_{\mathrm{sm}}(L) \neq \varnothing$.*

Proof. Choose $p \in X(L)$. Its multiplicity on the hypersurface X_L is one, two, or three. Multiplicity one means that p is smooth.

Suppose the multiplicity is two. In coordinates over L with $p = [1 : 0 : 0 : 0]$, the equation is

$$F = x_0 Q(x_1, x_2, x_3) + C(x_1, x_2, x_3), \qquad Q \neq 0,$$

where Q and C have degrees two and three, respectively. There exists $a \in L^3$ with $Q(a) \neq 0$, even over a finite field: the values $Q(e_i)$ and $Q(e_i + e_j) - Q(e_i) - Q(e_j)$ recover all coefficients of Q. The condition $Q(a) \neq 0$ ensures that the line joining p to $[0 : a_1 : a_2 : a_3]$ is not contained in X_L, and hence not in any plane component of X_L through p. The point

$$q = \left[-\frac{C(a)}{Q(a)} : a_1 : a_2 : a_3 \right]$$

lies on X_L and satisfies $\partial F / \partial x_0(q) = Q(a) \neq 0$. It is therefore a smooth point of X_L. In fact, this is the usual construction arising from projection from a double point; see [Kol02, Section 5].

Suppose now that p has multiplicity three. A vertex of a cubic cone has multiplicity three and is therefore covered by this case. In coordinates with $p = [1 : 0 : 0 : 0]$, the equation is independent of x_0. In the original coordinates, a nonzero vector $v \in L^4$ representing p consequently satisfies

$$F(x + sv) = F(x) \quad \text{in } L[x_0, x_1, x_2, x_3, s]. \tag{6.1}$$

If $\operatorname{char} k \neq 3$, consider the k-linear map

$$D_F : k^4 \longrightarrow k[x_0, x_1, x_2, x_3]_2, \qquad w \longmapsto D_w F = \sum_{i=0}^{3} w_i \frac{\partial F}{\partial x_i}.$$

Its kernel becomes nonzero over L, hence contains a nonzero vector $w \in k^4$. Euler's identity gives $0 = (D_w F)(w) = 3F(w)$, so $[w] \in X(k)$.

If $\operatorname{char} k = 3$, the hypothesis on $[L : k]$ implies that L/k is separable. Normalize v so that its i_0-th coordinate is 1 for some fixed i_0. Every conjugate $\sigma(v)$ satisfies (6.1), and directions satisfying this identity are closed under addition, since translations compose. Thus

$$w = \operatorname{Tr}_{L/k}(v) = \sum_{\sigma : L \hookrightarrow \overline{k}} \sigma(v) \in k^4$$

also satisfies (6.1). By our normalization, the i_0-th coordinate of w is $[L : k] \neq 0$ in k, so $w \neq 0$. Setting $x = 0$ and $s = 1$ in (6.1) then gives $F(w) = 0$. Finally, smoothness commutes with field extension, so the smooth points constructed on X_L belong to $X_{\mathrm{sm}}(L)$. □

Proposition 6.3. *Suppose every smooth cubic surface over $k((t))$ with a zero-cycle of degree one has a $k((t))$-rational point. Then every cubic surface over k with a zero-cycle of degree one has a k-rational point.*

Proof. Choose a closed point of degree d prime to three from the given zero-cycle, and let L/k be its residue field extension. By Lemma 6.2, we may assume that there is a point $q \in X_{\mathrm{sm}}(L)$.

Choose a homogeneous cubic G defining a smooth surface over k. For example, one may take

$$G = \begin{cases} x_0^3 + x_1^3 + x_2^3 + x_3^3, & \operatorname{char} k \neq 3, \\ x_0^3 + x_0 x_1^2 + x_1 x_2^2 + x_2 x_3^2, & \operatorname{char} k = 3. \end{cases}$$

Put $R = k[[t]]$, $K = k((t))$, and

$$\mathscr{X} = V\big((1 - t)F + tG\big) \subset \mathbb{P}^3_R.$$

This is a flat projective family with special fiber X and smooth generic fiber $Y = \mathscr{X}_K$. Indeed, the image in $\mathbb{A}^1_k$ of the closed locus defined by the pencil and its partial derivatives is closed by properness and excludes $t = 1$, so it excludes the generic point.

The point q lifts to a point $\tilde{q} \in \mathscr{X}(L[[t]])$ by Hensel's lemma: on an affine chart containing q, choose a coordinate for which the partial derivative of the defining equation is nonzero at q. Fixing the other coordinates at their values at q, Hensel's lemma lifts the remaining coordinate to a power series in t satisfying the equation of $\mathscr{X}$. (Alternatively, we may apply the smooth-point lifting statement of [Ma21, Lemma 2.4] to construct $\tilde{q}$.) The generic point of $\tilde{q}$ belongs to $Y(L((t)))$. A k-basis of L is also a K-basis of $M = L((t))$, so $[M : K] = d$, without any separability assumption. If y denotes the image in Y, the pushforward cycle is

$$\zeta = [M : \kappa(y)]\,[y], \qquad \deg_K \zeta = d.$$

Combining ζ with the degree-three intersection class with a line gives a zero-cycle of degree one on Y. Since Y is smooth and has a zero-cycle of degree one, the hypothesis of the proposition implies that $Y(K) \neq \varnothing$. Properness of $\mathscr{X}/R$ extends any point in $Y(K)$ to an R-point of $\mathscr{X}$, whose reduction belongs to $X(k)$. □

Proof of Theorem 6.1. Apply Proposition 6.3, using Theorem 5.3 over $k((t))$. □

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